Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy under stated assumptions.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pendulum” and “potential energy” to compare your model or result. Sentence starters are optional.

pendulum: an object that swings about a fixed support

potential energy: energy associated with the configuration of interacting objects

kinetic energy: energy associated with motion

01 Warm-up: bring back an earlier idea

With GMm=105 J·m, move an object from r=3 to r=7 m. **Find** work by gravity using U=-GMm/r.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mg Δh = ½mv²; v = √(2 g Δh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest 0.8 m above its lowest point reaches speed √(2 gh)=4 m/s there with g=10 and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

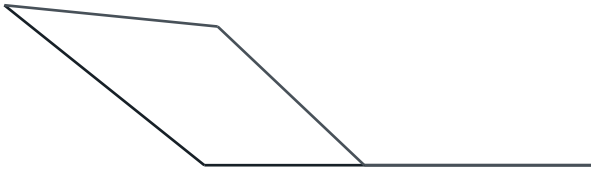
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two frictionless tracks start and end at the same heights but take different paths.
Compare final speeds from rest.

Sentence starter: Both ... . They differ in ... because ... .

Same start



Same finish

Mark the vertical height change.

Marzano check - circle: ① ② ③

04 Apply the physics

A coaster enters height 8 m at speed 6 m/s.
Find speed at height 3 m with no losses.
Use g=10.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A classmate uses total ramp length as h in mgh.
Explain how to identify the correct measurement.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

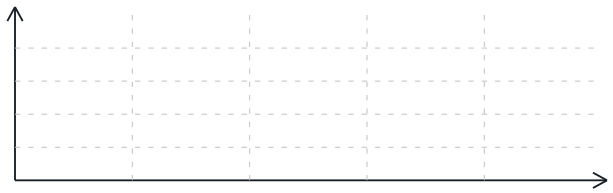
06 Review: connect an earlier idea

A positive triangular pulse peaks at 60 N and lasts 0.4 s. It acts on a 4 kg cart initially at -3 m/s.
Find impulse and final velocity.
Show the signed impulse on the force-time model.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Force (N)



Time (s)

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A cart begins at rest at height 5 m. Can it reach height 6 m without an external energy input? Justify.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A frictionless coaster starts from rest at height 8 m above the bottom of a vertical loop of radius 2 m.
Use g=10.
Find speed and normal force per unit mass at the top; check contact.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3